Chapter 7. Multivariate Time Series

VARMA models, inference, multivariate GARCH

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Many applications involve several series observed together. A portfolio, for instance, depends on each asset and on how the assets move jointly, and risk measures such as Value-at-Risk for a portfolio require a multivariate model. This chapter introduces multivariate time series, their stationarity, and the basic models: the vector ARMA (in particular VAR) model, cointegration, multivariate GARCH and copula-based models.

7.1 Multivariate time series models

Motivation. Stacking k univariate series into a vector, the notions of stationarity, autocovariance and ARMA modelling carry over almost verbatim, with matrices in place of scalars; the new phenomena are cross-dependence and cointegration.

Given k univariate series $Y_{1t}, \dots, Y_{kt}$, the vector $\mathbf{Y}_t = (Y_{1t}, \dots, Y_{kt})^T$ is a *k-dimensional multivariate time series*. For vectors $\mathbf{x} = (x_1, \dots, x_k)^T$ and $\mathbf{y} = (y_1, \dots, y_k)^T$, $\mathbf{x} \le \mathbf{y}$ means $x_i \le y_i$ for all i.

Definition 7.1.1 (Strict stationarity)

A multivariate series $\{\mathbf{Y}_t\}$ is *strictly stationary* if for all integers $t_1, \dots, t_m$ and h ($m \ge 1$) and all $\mathbf{y}_1, \dots, \mathbf{y}_m \in \mathbb{R}^k$,

$$P(\mathbf{Y}_{t_1} \le \mathbf{y}_1, \dots, \mathbf{Y}_{t_m} \le \mathbf{y}_m) = P(\mathbf{Y}_{t_1+h} \le \mathbf{y}_1, \dots, \mathbf{Y}_{t_m+h} \le \mathbf{y}_m).$$

Definition 7.1.2 (Stationarity)

A multivariate series $\{\mathbf{Y}_t\}$ is *stationary* if
(1) $E\mathbf{Y}_t = \boldsymbol{\mu}$ for all t;
(2) $EY_{it}^2 < \infty$ for all $1 \le i \le k$ and all t;
(3) $\operatorname{Cov}(Y_{it}, Y_{j,t+h}) = \operatorname{Cov}(Y_{i0}, Y_{jh})$ for all $1 \le i, j \le k$ and all t, h.

Definition 7.1.3 (Autocovariance function)

For a stationary $\{\mathbf{Y}_t\}$, the $k \times k$ matrix-valued function $\Gamma(\cdot) = (\Gamma_{ij}(\cdot))_{i,j=1}^k$ with

$$\Gamma_{ij}(h) = \operatorname{Cov}(Y_{it}, Y_{j,t+h}), \qquad h = 0, \pm 1, \pm 2, \dots,$$

is the *autocovariance function*; that is, $\Gamma(h) = E(\mathbf{Y}_t - \boldsymbol{\mu})(\mathbf{Y}_{t+h} - \boldsymbol{\mu})^T$ with $\boldsymbol{\mu} = E\mathbf{Y}_1$.

Theorem 7.1.4 (Properties of the autocovariance function)

The autocovariance function of a stationary series satisfies:
(1) $\Gamma(-h) = \Gamma(h)^T$;
(2) $|\Gamma_{ij}(h)| \le \sqrt{\Gamma_{ii}(0)}\sqrt{\Gamma_{jj}(0)}$;
(3) for all $n \ge 1$ and $\mathbf{a}_1, \dots, \mathbf{a}_n \in \mathbb{R}^k$, $\sum_{i,j=1}^n \mathbf{a}_i^T \Gamma(i-j)\mathbf{a}_j \ge 0$.

Proof. [Supplement] (The book leaves this as an exercise.)

(1) By stationarity, $\Gamma(-h) = E(\mathbf{Y}_t - \boldsymbol{\mu})(\mathbf{Y}_{t-h} - \boldsymbol{\mu})^T = E(\mathbf{Y}_{s+h} - \boldsymbol{\mu})(\mathbf{Y}_s - \boldsymbol{\mu})^T$ with $s = t - h$, and this is the transpose of $E(\mathbf{Y}_s - \boldsymbol{\mu})(\mathbf{Y}_{s+h} - \boldsymbol{\mu})^T = \Gamma(h)$.

(2) By the Cauchy–Schwarz inequality, $|\operatorname{Cov}(Y_{it}, Y_{j,t+h})| \le \sqrt{\operatorname{Var} Y_{it}}\sqrt{\operatorname{Var} Y_{j,t+h}} = \sqrt{\Gamma_{ii}(0)\Gamma_{jj}(0)}$.

(3) Since $\operatorname{Cov}(\mathbf{a}_i^T\mathbf{Y}_i, \mathbf{a}_j^T\mathbf{Y}_j) = \mathbf{a}_i^T\Gamma(j-i)\mathbf{a}_j$,

$$0 \le \operatorname{Var}\Big(\sum_{i=1}^{n} \mathbf{a}_i^T\mathbf{Y}_i\Big) = \sum_{i,j=1}^{n} \mathbf{a}_i^T\Gamma(j-i)\mathbf{a}_j = \sum_{i,j=1}^{n} \mathbf{a}_j^T\Gamma(i-j)\mathbf{a}_i,$$

where the last step transposes each scalar term and uses (1). Relabelling $i \leftrightarrow j$ gives the claim. ∎

Definition 7.1.5 (Vector ARMA model)

If $\{\mathbf{Y}_t\}$ satisfies

$$\mathbf{Y}_t - \Phi_1\mathbf{Y}_{t-1} - \cdots - \Phi_p\mathbf{Y}_{t-p} = \boldsymbol{\epsilon}_t + \Theta_1\boldsymbol{\epsilon}_{t-1} + \cdots + \Theta_q\boldsymbol{\epsilon}_{t-q}, \qquad \{\boldsymbol{\epsilon}_t\} \sim IID(\mathbf{0}, \Sigma), \tag{7.1}$$

where $\Phi_1, \ldots, \Phi_p, \Theta_1, \ldots, \Theta_q, \Sigma$ are $k \times k$ matrices and Σ is positive definite, then $\{\mathbf{Y}_t\}$ follows a *vector ARMA(p, q) model* (VARMA(p, q)). With $q = 0$ it is a *VAR(p) model*.

Write $\Phi(z) = I - \Phi_1 z - \cdots - \Phi_p z^p$ and $\Theta(z) = I + \Theta_1 z + \cdots + \Theta_q z^q$, so (7.1) reads $\Phi(B)\mathbf{Y}_t = \Theta(B)\boldsymbol{\epsilon}_t$. The next theorem is the analogue of the univariate causality result: the condition on the zeros of $\phi(z)$ becomes a condition on the zeros of $\det \Phi(z)$.

Theorem 7.1.6 (Causal VARMA)

If

$$\det(I - \Phi_1 z - \cdots - \Phi_p z^p) \neq 0 \qquad \text{for all complex } |z| \le 1,$$

then the VARMA model (7.1) has a stationary solution $\{\mathbf{Y}_t\}$, it is unique, and it has the representation

$$\mathbf{Y}_t = \sum_{i=0}^{\infty} C_i \boldsymbol{\epsilon}_{t-i}, \tag{7.2}$$

where $C_0 = I_k$ and the $k \times k$ matrices C_i are determined by

$$(I - \Phi_1 z - \cdots - \Phi_p z^p) \sum_{i=0}^{\infty} C_i z^i = I + \Theta_1 z + \cdots + \Theta_q z^q, \qquad |z| \le 1. \tag{7.3}$$

With mean $\boldsymbol{\mu}$ the model reads $\mathbf{Y}_t - \boldsymbol{\mu} - \Phi_1(\mathbf{Y}_{t-1} - \boldsymbol{\mu}) - \cdots - \Phi_p(\mathbf{Y}_{t-p} - \boldsymbol{\mu}) = \boldsymbol{\epsilon}_t + \Theta_1\boldsymbol{\epsilon}_{t-1} + \cdots + \Theta_q\boldsymbol{\epsilon}_{t-q}$.

Proof. [Supplement] (The book omits the proof.) Matrix products do not commute, so the order of factors in (7.3) matters: $\Phi(z)$ stands on the left.

Step 1: the coefficients. $\det \Phi(z)$ is a polynomial with finitely many zeros, none in the closed unit disc, so it has no zeros in $|z| < 1 + \delta$ for some $\delta > 0$. Hence $\Phi(z)^{-1} = \operatorname{adj}\Phi(z)/\det\Phi(z)$ has entries that are rational functions without poles in $|z| < 1 + \delta$, i.e. analytic there, and so is $C(z) := \Phi(z)^{-1}\Theta(z) = \sum_{i \ge 0} C_i z^i$. The power series converges at $z = 1 + \delta/2$, so its terms are bounded: $\|C_i\| \le K(1 + \delta/2)^{-i}$ for a constant K. In particular $\sum_i \|C_i\| < \infty$, $C_0 = C(0) = I$, and $\Phi(z)C(z) = \Theta(z)$, which is (7.3). Similarly $\Phi(z)^{-1} = \sum_i D_i z^i$ with $\sum_i \|D_i\| < \infty$.

Step 2: existence. Since $E\|\boldsymbol{\epsilon}_t\| \le (\operatorname{tr}\Sigma)^{1/2}$, $E\sum_i \|C_i\|\|\boldsymbol{\epsilon}_{t-i}\| < \infty$, so the series (7.2) converges absolutely a.s.; it also converges in mean square because $\big(E\|\sum_{i=m}^{n} C_i\boldsymbol{\epsilon}_{t-i}\|^2\big)^{1/2} \le (\operatorname{tr}\Sigma)^{1/2}\sum_{i=m}^{n}\|C_i\| \to 0$. Then $E\mathbf{Y}_t = \mathbf{0}$ and, as $E\boldsymbol{\epsilon}_s\boldsymbol{\epsilon}_r^T = \Sigma\,\mathbb{1}\{s = r\}$,

$$E\mathbf{Y}_t\mathbf{Y}_{t+h}^T = \sum_{i,j} C_i E(\boldsymbol{\epsilon}_{t-i}\boldsymbol{\epsilon}_{t+h-j}^T)C_j^T = \sum_{i=0}^{\infty} C_i \Sigma C_{i+h}^T \qquad (h \ge 0),$$

which does not depend on t; so $\{\mathbf{Y}_t\}$ is stationary. Writing $\Phi(z)=\sum_{j=0}^{p}A_j z^j$ ($A_0=I$, $A_j=-\Phi_j$) and rearranging the absolutely convergent double series,

$$\Phi(B)\mathbf{Y}_t=\sum_{j=0}^{p}A_j\sum_{i\ge 0}C_i\boldsymbol{\epsilon}_{t-j-i}=\sum_{m\ge 0}\Big(\sum_{j=0}^{\min(p,m)}A_jC_{m-j}\Big)\boldsymbol{\epsilon}_{t-m}=\sum_{m=0}^{q}\Theta_m\boldsymbol{\epsilon}_{t-m},$$

because the inner sum is the coefficient of z^m in $\Phi(z)C(z)=\Theta(z)$ ($\Theta_0=I$). Thus (7.2) solves (7.1).

Step 3: uniqueness. Let $\{\mathbf{Y}_t\}$ be any stationary solution. Applying the filter $\sum_i D_iB^i$ (with summable coefficients, so it acts in mean square on stationary sequences and composes like multiplication of power series) to $\Phi(B)\mathbf{Y}_t=\Theta(B)\boldsymbol{\epsilon}_t$ gives $\mathbf{Y}_t=\Phi(B)^{-1}\Theta(B)\boldsymbol{\epsilon}_t=\sum_i C_i\boldsymbol{\epsilon}_{t-i}$. ■

Example 7.1.7 (Coefficients of a bivariate ARMA(1,1))

Let $k=2$, $p=q=1$, $\Phi_1=\begin{pmatrix}0.5 & 0.5\\ 0 & 0.5\end{pmatrix}$ and $\Theta_1=\Phi_1^T$. Here $\det\Phi(z)=(1-0.5z)^2\neq 0$ for $|z|\le 1$. From (7.3),

$$\begin{aligned}\sum_{i=0}^{\infty}C_iz^i&=\begin{pmatrix}1-0.5z & -0.5z\\ 0 & 1-0.5z\end{pmatrix}^{-1}\begin{pmatrix}1+0.5z & 0\\ 0.5z & 1+0.5z\end{pmatrix}\\&=(1-0.5z)^{-2}\begin{pmatrix}1 & 0.5z(1+0.5z)\\ 0.5z(1-0.5z) & 1-0.25z^2\end{pmatrix}.\end{aligned}$$

Expanding $(1-0.5z)^{-2}=\sum_{m\ge 0}(m+1)2^{-m}z^m$ and collecting the coefficient of z^i entrywise,

$$C_0=I,\qquad C_i=2^{-i}\begin{pmatrix}i+1 & 2i-1\\ 1 & 2\end{pmatrix},\quad i=1,2,\dots$$

For instance the $(1,2)$ entry is $\frac{1}{2}\,i\,2^{-(i-1)}+\frac{1}{4}(i-1)2^{-(i-2)}=(2i-1)2^{-i}$.

7.1.1 Cointegration

Definition 7.1.8 (Integration order and cointegration)

If $(1-B)^d\mathbf{Y}_t=\big((1-B)^dY_{1t},\dots,(1-B)^dY_{kt}\big)^T$ is stationary while $\mathbf{Y}_t,(1-B)\mathbf{Y}_t,\dots,(1-B)^{d-1}\mathbf{Y}_t$ are nonstationary, we write $\mathbf{Y}_t\sim I(d)$. If $\mathbf{Y}_t\sim I(d)$ and, for some positive integer $\delta<d$ and some vector $\boldsymbol{\alpha}\in\mathbb{R}^k$, $\boldsymbol{\alpha}^T\mathbf{Y}_t\sim I(\delta)$, then $\mathbf{Y}_t$ is *cointegrated* with *cointegrating vector* $\boldsymbol{\alpha}$. (The most common case is $d=1$, $\delta=0$: a stationary combination of $I(1)$ series.)

Cointegration must be distinguished from *spurious regression*. If X_t and Y_t are two *independent* random walks, there is no relation between them, yet regressing Y_t on X_t typically produces a large R^2 and a highly significant slope, because both series wander with apparent trends. Cointegration, in contrast, is a genuine long-run equilibrium relation.

Example 7.1.10 (Consumption and income)

Let C_t be real per-capita consumption and I_t real per-capita disposable income, with $C_t,I_t\sim I(1)$, and suppose $C_t=a+bI_t+u_t$ for some $a,b\in(0,1)$ with $Eu_t=0$, $u_t\sim I(0)$. Then $\boldsymbol{\alpha}^T\mathbf{Y}_t=C_t-bI_t=a+u_t\sim I(0)$ for $\mathbf{Y}_t=(C_t,I_t)^T$ and $\boldsymbol{\alpha}=(1,-b)^T$: the pair is cointegrated.

When the components are cointegrated, the *vector error correction model* (VECM) is the natural form.

Definition 7.1.11 (VECM)

The VECM(p) with *cointegration rank* r ($\le k$) is

$$\Delta \mathbf{Y}_t = \boldsymbol{\delta} + \Pi \mathbf{Y}_{t-1} + \sum_{i=1}^{p-1} \Phi_i^* \Delta \mathbf{Y}_{t-i} + \boldsymbol{\epsilon}_t, \qquad \{\boldsymbol{\epsilon}_t\} \sim IID(\mathbf{0}, \Sigma), \tag{7.4}$$

where $\Delta = 1 - B$ is the differencing operator, $\boldsymbol{\delta}$ is a $k \times 1$ constant vector, $\Pi = \boldsymbol{\alpha}\boldsymbol{\beta}^T$ with $\boldsymbol{\alpha}, \boldsymbol{\beta}$ $k \times r$ matrices, and the Φ_i^* are $k \times k$ matrices.

If $\mathbf{Y}_t \sim I(1)$, then $\Delta \mathbf{Y}_t$ and $\boldsymbol{\epsilon}_t$ are stationary, so (7.4) forces $\boldsymbol{\alpha}\boldsymbol{\beta}^T \mathbf{Y}_{t-1}$, and hence (when $\boldsymbol{\alpha}$ has full column rank) $\boldsymbol{\beta}^T \mathbf{Y}_{t-1}$, to be stationary: the columns of $\boldsymbol{\beta}$ are r cointegrating vectors, and $\boldsymbol{\alpha}$ measures how fast deviations from equilibrium are corrected.

Johansen (1988) proposed a test of $H_0 : r \le r_0$ against $H_1 : r > r_0$ based on the *trace statistic*

$$\lambda_{\text{trace}}(r_0) = -n \sum_{i=r_0+1}^{k} \log(1 - \hat{\lambda}_i), \tag{7.5}$$

where n is the sample size and $\hat{\lambda}_1 \ge \cdots \ge \hat{\lambda}_k$ are the estimated eigenvalues associated with Π. If $r \le r_0$, the true eigenvalues beyond r_0 vanish and the statistic stays small; large values reject H_0. Its null distribution is nonstandard and depends on the constant term, r_0, trend terms and the dimension (see Johansen, 1988).

7.2 Inference for stationary multivariate series

Motivation. The sample mean and sample autocovariance matrices are consistent estimators, exactly as in the univariate case, and a VAR(p) model is fitted by least squares one equation at a time.

Theorem 7.2.1 (Sample mean)

Let $\mathbf{Y}_t = \boldsymbol{\mu} + \sum_{i=0}^{\infty} C_i \boldsymbol{\epsilon}_{t-i}$ with $\{\boldsymbol{\epsilon}_t\} \sim IID(\mathbf{0}, \Sigma)$ and $k \times k$ matrices C_i satisfying $\sum_{i=0}^{\infty} \|C_i\| < \infty$, where $\|C\| = \sup\{\|C\mathbf{y}\| : \|\mathbf{y}\| \le 1\}$. Let $\hat{\boldsymbol{\mu}} = \frac{1}{n}\sum_{t=1}^{n} \mathbf{Y}_t$. Then:
(1) $P(\hat{\boldsymbol{\mu}} \to \boldsymbol{\mu},\ n \to \infty) = 1$;
(2) if $\Gamma_{ii}(n) \to 0$ for all $1 \le i \le k$, then $E(\hat{\boldsymbol{\mu}} - \boldsymbol{\mu})^T(\hat{\boldsymbol{\mu}} - \boldsymbol{\mu}) \to 0$;
(3) if $\sum_{h=0}^{\infty} |\Gamma_{ii}(h)| < \infty$ for all $1 \le i \le k$, then

$$n E(\hat{\boldsymbol{\mu}} - \boldsymbol{\mu})^T(\hat{\boldsymbol{\mu}} - \boldsymbol{\mu}) \to \sum_{i=1}^{k} \sum_{h=-\infty}^{\infty} \Gamma_{ii}(h).$$

Proof. [Supplement] (The book omits the proof.) By Theorem 7.1.6's Step 2 argument, the series converges a.s. and in mean square and $\{\mathbf{Y}_t\}$ is stationary.

(1) $\mathbf{Y}_t = G(\boldsymbol{\epsilon}_t, \boldsymbol{\epsilon}_{t-1}, \dots)$ for one fixed measurable G, so $\{\mathbf{Y}_t\}$ is a strictly stationary ergodic sequence (a function of an i.i.d. shift), and $E\|\mathbf{Y}_t\| < \infty$. Birkhoff's ergodic theorem gives $\hat{\boldsymbol{\mu}} \to \boldsymbol{\mu}$ a.s.

(2) $E(\hat{\boldsymbol{\mu}} - \boldsymbol{\mu})^T(\hat{\boldsymbol{\mu}} - \boldsymbol{\mu}) = \sum_{i=1}^{k} \mathrm{Var}(\bar{Y}_i)$ and

$$\mathrm{Var}(\bar{Y}_i) = \frac{1}{n^2} \sum_{s,t=1}^{n} \Gamma_{ii}(t-s) = \frac{1}{n} \sum_{|h|<n} \Big(1 - \frac{|h|}{n}\Big) \Gamma_{ii}(h) \le \frac{1}{n} \sum_{|h|<n} |\Gamma_{ii}(h)|.$$

If $\Gamma_{ii}(h) \to 0$ then its Cesàro means $\frac{1}{n}\sum_{|h|<n} |\Gamma_{ii}(h)|$ tend to 0.

(3) $n\,\mathrm{Var}(\bar{Y}_i) = \sum_{|h|<n}(1-|h|/n)\Gamma_{ii}(h) \to \sum_h \Gamma_{ii}(h)$ by dominated convergence (dominating sequence $|\Gamma_{ii}(h)|$, summable by assumption, using $\Gamma_{ii}(-h)=\Gamma_{ii}(h)$). Summing over i gives the claim. ■

Theorem 7.2.2 (Sample autocovariance)

Let $\{\mathbf{Y}_t\}$ be as in Theorem 7.2.1. Then $\Gamma(h) = \sum_{i=0}^{\infty} C_i \Sigma C_{i+h}^T$ $(h \geq 0)$, and the sample autocovariance function

$$\hat{\Gamma}(h) = \frac{1}{n}\sum_{t=1}^{n-|h|}(\mathbf{Y}_t - \hat{\boldsymbol{\mu}})(\mathbf{Y}_{t+|h|} - \hat{\boldsymbol{\mu}})^T$$

(for $h<0$ take the transpose) satisfies $P(\hat{\Gamma}(h) \to \Gamma(h),\ n\to\infty) = 1$; that is, $\hat{\Gamma}(\cdot)$ is a consistent estimator of $\Gamma(\cdot)$.

Proof. [Supplement] The formula for $\Gamma(h)$ was derived in Step 2 of the proof of Theorem 7.1.6. Fix $h \geq 0$ and write $\mathbf{Y}_t - \hat{\boldsymbol{\mu}} = (\mathbf{Y}_t - \boldsymbol{\mu}) - (\hat{\boldsymbol{\mu}} - \boldsymbol{\mu})$:

$$\begin{aligned}\hat{\Gamma}(h) = \frac{1}{n}\sum_{t=1}^{n-h}(\mathbf{Y}_t-\boldsymbol{\mu})(\mathbf{Y}_{t+h}-\boldsymbol{\mu})^T - \Big(\frac{1}{n}\sum_{t=1}^{n-h}(\mathbf{Y}_t-\boldsymbol{\mu})\Big)(\hat{\boldsymbol{\mu}}-\boldsymbol{\mu})^T \\ - (\hat{\boldsymbol{\mu}}-\boldsymbol{\mu})\Big(\frac{1}{n}\sum_{t=1}^{n-h}(\mathbf{Y}_{t+h}-\boldsymbol{\mu})\Big)^T + \frac{n-h}{n}(\hat{\boldsymbol{\mu}}-\boldsymbol{\mu})(\hat{\boldsymbol{\mu}}-\boldsymbol{\mu})^T.\end{aligned}$$

The matrices $Z_t = (\mathbf{Y}_t-\boldsymbol{\mu})(\mathbf{Y}_{t+h}-\boldsymbol{\mu})^T$ are a fixed function of the shifted i.i.d. sequence and integrable (Cauchy–Schwarz), so the ergodic theorem gives $\frac{1}{n}\sum_{t\leq n-h} Z_t \to EZ_t = \Gamma(h)$ a.s. (dropping h terms changes nothing in the limit). The averages in the middle terms tend to $\mathbf{0}$ a.s. and $\hat{\boldsymbol{\mu}} - \boldsymbol{\mu} \to \mathbf{0}$ a.s. by Theorem 7.2.1(1), so the other three terms vanish in the limit. ■

7.2.1 Inference in VAR(p) models

In the VAR(p) model $\mathbf{Y}_t - \Phi_1\mathbf{Y}_{t-1} - \cdots - \Phi_p\mathbf{Y}_{t-p} = \boldsymbol{\epsilon}_t$, $\{\boldsymbol{\epsilon}_t\} \sim IID(\mathbf{0},\Sigma)$, the coefficients are estimated by least squares, given initial values $\mathbf{Y}_{-p+1},\ldots,\mathbf{Y}_0$. Define

$$\mathbf{y}_t = \begin{pmatrix}\mathbf{Y}_t\\ \vdots \\ \mathbf{Y}_{t-p+1}\end{pmatrix}\ (kp\times 1), \qquad \begin{aligned} Y &= (\mathbf{Y}_1,\ldots,\mathbf{Y}_n)\ (k\times n), & \Phi &= (\Phi_1,\ldots,\Phi_p)\ (k\times kp),\\ X &= (\mathbf{y}_0,\ldots,\mathbf{y}_{n-1})\ (kp\times n), & \epsilon &= (\boldsymbol{\epsilon}_1,\ldots,\boldsymbol{\epsilon}_n)\ (k\times n).\end{aligned}$$

Then the model is $Y = \Phi X + \epsilon$.

Proposition (Least squares for VAR(p), [Supplement])

If XX^T is invertible, then for every Φ

$$(Y-\Phi X)(Y-\Phi X)^T = (Y-\hat{\Phi}X)(Y-\hat{\Phi}X)^T + (\hat{\Phi}-\Phi)XX^T(\hat{\Phi}-\Phi)^T, \qquad \hat{\Phi} = YX^T(XX^T)^{-1}.$$

Hence $\hat{\Phi}$ minimizes $\sum_{t=1}^n \|\mathbf{Y}_t - \Phi\mathbf{y}_{t-1}\|^2 = \mathrm{tr}\{(Y-\Phi X)(Y-\Phi X)^T\}$ (and the determinant of the residual cross-product matrix), uniquely.

Proof. Write $Y-\Phi X = (Y-\hat{\Phi}X) + (\hat{\Phi}-\Phi)X$ and expand. The cross terms vanish because $(Y-\hat{\Phi}X)X^T = YX^T - YX^T(XX^T)^{-1}XX^T = 0$. The second term on the right is positive semidefinite, and it is zero only if $\Phi = \hat{\Phi}$ since XX^T is positive definite. Taking traces (or determinants, using monotonicity of det on positive semidefinite matrices) gives the minimization claims. ■

The least squares estimators are therefore

$$\hat{\Phi} = YX^T(XX^T)^{-1}, \qquad \hat{\Sigma} = \frac{1}{n-kp}(Y-\hat{\Phi}X)(Y-\hat{\Phi}X)^T;$$

note that $\hat{\Phi}$ is the same as ordinary least squares applied separately to each of the k equations.

The order p can be chosen as follows. For a sufficiently large p_0, fit VAR(i), $i = 1, \dots, p_0$, compute the residuals $\hat{\boldsymbol{\epsilon}}_t^{(i)} = \mathbf{Y}_t - \hat{\Phi}_1^{(i)}\mathbf{Y}_{t-1} - \cdots - \hat{\Phi}_i^{(i)}\mathbf{Y}_{t-i}$ and their covariance matrix

$$\hat{\Sigma}_i = \frac{1}{n-2i}\sum_{t=i+1}^{n} \hat{\boldsymbol{\epsilon}}_t^{(i)}\hat{\boldsymbol{\epsilon}}_t^{(i)T},$$

and select the i minimizing

$$AIC(i) = \log|\hat{\Sigma}_i| + 2k^2 i/n \qquad \text{or} \qquad BIC(i) = \log|\hat{\Sigma}_i| + k^2 i \log(n)/n.$$

The penalty counts the $k^2 i$ autoregressive parameters.

7.3 Multivariate GARCH models

Motivation. The volatility of a portfolio depends on the joint, time-varying covariance of its assets, so the univariate GARCH idea must be extended to a conditional covariance matrix.

Let y_{it}, $i = 1, \dots, d$, be d series forming $\mathbf{Y}_t$, and let $\mathcal{F}_{t-1}$ be the information up to $t-1$. With $\sigma_{iit} = \mathrm{Var}(y_{it} \mid \mathcal{F}_{t-1})$ and $\sigma_{ijt} = \mathrm{Cov}(y_{it}, y_{jt} \mid \mathcal{F}_{t-1})$, the matrix $\Sigma_t = (\sigma_{ijt})$ is the conditional covariance matrix of $\mathbf{Y}_t$. The simplest multivariate GARCH model is

$$\mathbf{Y}_t = \Sigma_t^{1/2}\boldsymbol{\epsilon}_t, \qquad \{\boldsymbol{\epsilon}_t\} \sim IID(\mathbf{0}, I_d), \qquad E(\mathbf{Y}_t\mathbf{Y}_t^T \mid \mathcal{F}_{t-1}) = \Sigma_t, \tag{7.6}$$

with Σ_t a function of the past. Since Σ_t has $d(d+1)/2$ distinct entries, an unrestricted recursion for Σ_t has a number of parameters growing like d^4, and estimation quickly becomes infeasible. Practical models therefore restrict the form of Σ_t; well-known examples are the BEKK (Baba–Engle–Kraft–Kroner) model, which keeps Σ_t positive definite by a quadratic-form recursion, and the CCC (constant conditional correlation) GARCH model, $\Sigma_t = D_t R D_t$ with $D_t = \mathrm{diag}(\sigma_{11t}^{1/2}, \dots, \sigma_{ddt}^{1/2})$, univariate GARCH volatilities and a constant correlation matrix R. See Bollerslev (1990) and Engle and Kroner (1995).

7.4 SCOMDY models

Motivation. A copula separates a joint distribution into its marginals and a dependence function. Combining univariate GARCH volatilities with a copula for the standardized errors gives a flexible, semiparametric multivariate model.

A simple instance of (7.6) consists of d GARCH(1,1) series whose errors are jointly normal:

$$y_{it} = \sigma_{it}\xi_{it}, \quad i = 1, \dots, d, \qquad \boldsymbol{\xi}_t = (\xi_{1t}, \dots, \xi_{dt})^T \text{ i.i.d. multivariate normal.} \tag{7.7}$$

If $\boldsymbol{\xi}_t$ is not normal, its distribution must be modelled. The *semiparametric copula-based multivariate dynamic* (SCOMDY) model does this with a copula.

Theorem (Sklar, continuous case, [Supplement])

Let $\mathbf{Y} = (Y_1, \dots, Y_d)^T$ have joint distribution function F and continuous marginals $F_1, \dots, F_d$. Then there is a function C on $[0,1]^d$, namely the distribution function of $(F_1(Y_1), \dots, F_d(Y_d))$, such that

$$F(\mathbf{y}) = C(F_1(y_1), \dots, F_d(y_d)), \qquad \mathbf{y} = (y_1, \dots, y_d)^T,$$

and C has uniform $(0,1)$ marginals (a *copula*).

Proof. If F_i is continuous then $U_i = F_i(Y_i)$ is uniform: for $u \in (0,1)$ let $x_u = \max\{x : F_i(x) = u\}$ (exists by continuity, as $F_i(x) \to 0, 1$ at $\mp\infty$); then $\{F_i(Y_i) \le u\} = \{Y_i \le x_u\}$ up to a null set, so $P(U_i \le u) = F_i(x_u) = u$. Moreover $\{Y_i \le y_i\} \subset \{F_i(Y_i) \le F_i(y_i)\}$, and the difference $\{Y_i > y_i,\ F_i(Y_i) = F_i(y_i)\}$ has probability 0 because F_i is flat on the corresponding interval. Hence

$$F(\mathbf{y}) = P(Y_i \le y_i\ \forall i) = P(U_i \le F_i(y_i)\ \forall i) = C(F_1(y_1), \dots, F_d(y_d)).$$

■

Thus the copula C carries all the dependence. Since parametric multivariate families other than the normal are scarce, one models C instead of F. Common choices are the Gaussian, Archimedean, Frank and Clayton copulas. For $d = 2$ the *Gaussian copula* is

$$C(u, v; \rho) = \Phi_\rho(\Phi^{-1}(u), \Phi^{-1}(v)),$$

where Φ is the standard normal distribution function and Φ_ρ the bivariate normal distribution function with means 0, variances 1 and correlation ρ. An *Archimedean copula* has the form

$$C(u, v) = \varphi^{-1}(\varphi(u) + \varphi(v))$$

for a convex generator φ on $[0,1]$ (decreasing, $\varphi(1) = 0$); $\varphi(t) = \log(e^{-\theta} - 1) - \log(e^{-\theta t} - 1)$, $\theta \neq 0$, gives the *Frank copula* and $\varphi(t) = (t^{-\theta} - 1)/\theta$, $\theta > 0$, the *Clayton copula*. As these examples show, copulas usually involve unknown parameters (see Joe, 1997).

Marginal distribution functions are estimated by empirical distribution functions. Given $\mathbf{Y}_j = (Y_{1j}, \dots, Y_{dj})^T$, $j = 1, \dots, n$, choose a copula family C and estimate F by $\hat{F}(\mathbf{y}) = C(\hat{F}_1(y_1), \dots, \hat{F}_d(y_d))$, where

$$\hat{F}_i(y) = \frac{1}{n} \sum_{j=1}^{n} I(Y_{ij} \le y), \qquad i = 1, \dots, d.$$

For a series following (7.7) with copula-distributed errors, this yields a conditional likelihood for $(y_{1t}, \dots, y_{dt})^T$, $t = 1, \dots, n$, on which inference is based. The SCOMDY model is very general: GARCH, VAR and Markov-switching models are special cases. Chen and Fan (2006) estimated the copula parameters by a Kullback–Leibler criterion, derived their asymptotic distribution, and proposed model-selection tests.